# EGOHUMANOID-V2: HUMAN-TO-HUMANOID TRANSFER OF COORDINATED WHOLE-BODY SKILLS FOR LOCO-MANIPULATION

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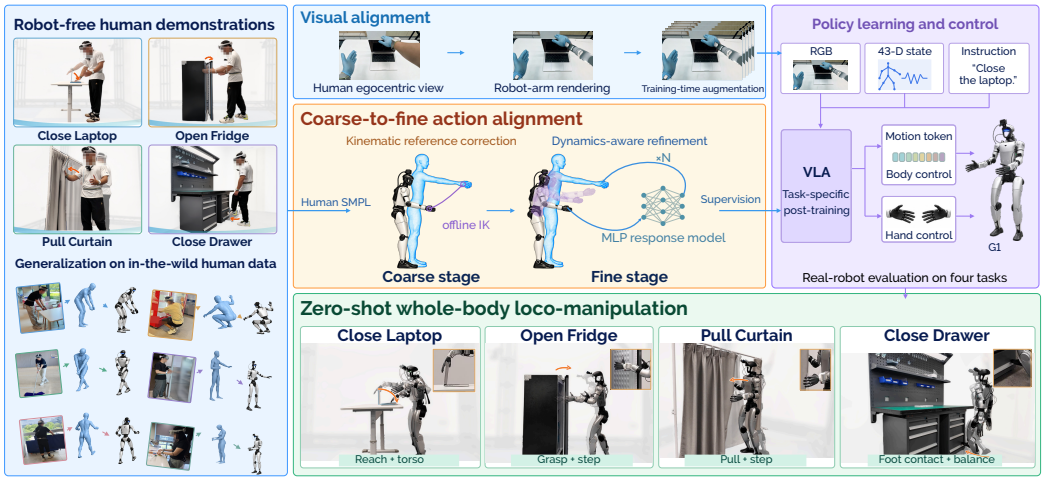



Figure 1: Introducing **EgoHumanoid-V2**, a human-to-humanoid skill transfer framework for whole-body loco-manipulation. Robot teleoperation is costly and hard to scale across environments, while human demonstrations capture diverse scenes and coordinated whole-body skills. Action and visual alignment bridge the embodiment gap for VLA training. Experiments on four real-world tasks show zero-shot skill transfer without target-task robot demonstrations.

## ABSTRACT

Human demonstrations capture diverse scenes and rich whole-body skills without requiring robot teleoperation. Prior work on egocentric transfer has emphasized scene generalization in loco-manipulation under decoupled control, leaving direct transfer of coordinated whole-body skills less explored. We present **EgoHumanoid-V2**, a human-to-humanoid framework to transfer coordinated whole-body loco-manipulation skills from egocentric demonstrations. At its core, coarse-to-fine action alignment combines kinematic reference correction with dynamics-aware refinement. It improves end-effector pose accuracy while preserving whole-body coordination. We also use robot-arm rendering and training-time image augmentation to reduce the visual embodiment gap and improve viewpoint robustness. On four real-world tasks, vision-language-action (VLA) policies trained on aligned human data show zero-shot skill transfer without target-task robot demonstrations. Task scores are comparable to those of policies trained on teleoperation data at a lower collection cost. These results support human data as direct skill supervision. Demos are available at egohumanoid-v2-iclr2027.github.io.

## 1 INTRODUCTION

Human demonstrations offer rich manipulation experience across diverse scenes without robot teleoperation (Grauman et al., 2022; 2024). Yet the embodiment gap makes it challenging to exploit human motion for robot policy learning. Some approaches learn loco-manipulation priors from videos without action labels (Ye et al., 2025; Bu et al., 2025; Jiang et al., 2026; Ye et al., 2026). This

bypasses explicit action alignment, but may underutilize human motion (Lin et al., 2026). Recent work captures human hand trajectories with the Universal Manipulation Interface (UMI) (Chi et al., 2024; Liu et al., 2026) or VR motion capture (Yuan et al., 2026; Shi et al., 2026). Relative end-effector pose changes provide a shared human–robot action space (Kareer et al., 2026; Yuan et al., 2026). However, these methods primarily transfer bimanual manipulation and navigation, rather than coordinated whole-body loco-manipulation skills.

Humans naturally coordinate their arms, torso, and legs to extend their workspace and manipulate efficiently. Transferring these coordinated skills, however, requires more than matching end-effector trajectories. Whole-body tracking policies enable such coordination through joint whole-body control (Luo et al., 2026; Pan et al., 2026b). Unlike decoupled controllers that support arm inverse kinematics (IK) alongside separate balance and locomotion control (Ben et al., 2025; Li et al., 2025a), these policies jointly generate commands from full-body feedback. However, tracking human references alone can leave substantial end-effector error (Table 1). A straightforward approach is to correct the tracker's joint commands using online IK. Yet these corrections override part of the coordinated output without adjusting the remaining commands, potentially disrupting balance. Our implementation and evaluation confirm this risk (Appendix D). The challenge is therefore to improve interaction accuracy while preserving whole-body coordination.

We present **EgoHumanoid-V2**, a human-to-humanoid framework for transferring coordinated whole-body loco-manipulation skills from egocentric demonstrations (Fig. 1). Building on EgoHumanoid's exploration of egocentric human-to-humanoid transfer (Shi et al., 2026), we extend the scope from loco-manipulation under decoupled control to coordinated whole-body skills. At its core is offline, coarse-to-fine action alignment that accounts for the tracking policy's execution. In the coarse stage, kinematic reference correction uses offline IK to align motion references with human hand or foot targets, reducing geometric mismatch before tracking. In the fine stage, dynamics-aware refinement combines rollout feedback with a learned local response model to iteratively improve end-effector tracking. Together, these stages improve manipulation accuracy while preserving whole-body coordination. We also use robot-arm rendering to reduce the visual embodiment gap and training-time image augmentation to improve viewpoint robustness. The resulting observations and actions provide paired supervision for vision-language-action (VLA) post-training.

On four real-world tasks, policies trained on 200 human demonstrations per task achieve a mean score of 51.67% without target-task robot demonstrations. On two tasks, human collection requires 76.36% less labor per demonstration than teleoperation. Ablations examine action alignment through tracking accuracy and motion quality, and visual alignment through policy performance.

Our contributions are: **(a)** A human-to-humanoid framework for transferring coordinated whole-body loco-manipulation skills from egocentric demonstrations. **(b)** Coarse-to-fine action alignment through kinematic reference correction and dynamics-aware refinement. **(c)** Evidence of zero-shot skill transfer on four real-world tasks, with analyses of accuracy, policy scores, and collection cost.

## 2 Related Work

### 2.1 Whole-Body Control for Loco-Manipulation

Humanoid whole-body control supports imitation (Ji et al., 2025; Fu et al., 2025) and teleoperation (He et al., 2024; 2025b; Ze et al., 2025). For loco-manipulation, decoupled controllers combine arm joint targets or IK with adaptive lower-body control (Ben et al., 2025; Li et al., 2025a). Whole-body tracking policies instead use motion references and full-body feedback for joint control (Chen et al., 2026; Pan et al., 2026b), supporting diverse motions (Luo et al., 2026; Qi et al., 2026) and precise spatial tracking (Xu et al., 2026a). Retargeting improves reference quality (Araujo et al., 2026), interaction geometry (Yang et al., 2026b), and dynamic feasibility (Pan et al., 2026a). Execution errors nevertheless remain. Residual learning addresses dynamics mismatch (He et al., 2025a) and task interactions (Zhao et al., 2026). We instead correct references and refine execution offline under SONIC, keeping the controller fixed across tasks.

### 2.2 Learning from Human Demonstrations

Human demonstrations support robot learning through visual representations (Nair et al., 2023; Ma et al., 2023; Zeng et al., 2024), predictive pretraining (Wu et al., 2024), and motion priors from latent action models (Ye et al., 2025; Bu et al., 2025; Jiang et al., 2026) or world action models (Ye et al., 2026). Captured human trajectories provide direct action supervision (Hoque et al., 2026) through UMI-based interfaces (Chi et al., 2024; Zhaxizhuoma et al., 2025; Xu et al., 2025; Liu et al., 2026) or VR capture (Yuan et al., 2026). Wearable hand capture enables human–robot co-training (Kareer et al., 2025; Qiu et al., 2025; Tao et al., 2025), with domain adaptation improving transfer (Punamiya et al., 2025). Recent egocentric systems study scaling (Zheng et al., 2026) and humanoid control (Wei et al., 2026; Yang et al., 2026a). Loco-manipulation transfer extends to wheeled robots (Xu et al., 2026b) and humanoids with decoupled control (Shi et al., 2026; Zhong et al., 2026). Scene reconstruction supports contextual whole-body imitation (Allshire et al., 2025), while HuMI and BifrostUMI transfer coordinated manipulation (Nai et al., 2026; Wang et al., 2026). We instead refine action supervision offline under a fixed tracker, without task-specific training or fine-tuning.

## 3 Method

Figure 3 illustrates our action and visual alignment pipelines.

### 3.1 Problem Setup

For each task $\tau$, human demonstrations contain synchronized egocentric images $I$, body motion $x$, and hand signals $h$, together with an instruction $\ell$. We convert them into robot observations and actions to post-train a VLA policy $\pi_\theta^\tau$ without target-task robot demonstrations. The policy predicts whole-body and hand actions from images, proprioceptive states, and instructions.

### 3.2 Data Collection and Policy Learning

We use a wearable system based on EgoHumanoid (Shi et al., 2026) to capture human demonstrations without robot hardware (Figure 2).

**Human demonstration capture.** A neck-mounted GoPro captures egocentric RGB images, while the PICO system captures body motion and hand poses. These streams are synchronized at 50 Hz. For deployment, we use a Unitree G1 with a head-mounted GoPro and BrainCo Revo2 hands. The human and robot cameras are placed at similar heights to reduce the viewpoint gap.

**Training data construction.** Offline action alignment converts human motion into SONIC tokens (Luo et al., 2026). Controller rollouts provide humanoid states for policy inputs and robot-arm rendering. The adapted images and aligned actions form training pairs, with image augmentation applied during training.

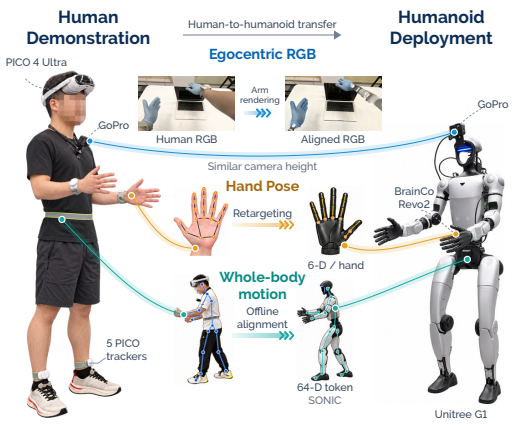



Figure 2: Hardware for human demonstration capture and humanoid deployment.

**Policy learning and deployment.** Using the aligned human demonstrations, we post-train one $\pi_{0.5}$ policy (Intelligence et al., 2025) per task without target-task robot demonstrations. The policy predicts action sequences from egocentric images, humanoid states, and language instructions, with a common interface for training and deployment:

$$o_t = (\widetilde{I}_t, q_t, \ell), \qquad a_t = (z_t, h_t^{\mathrm{L}}, h_t^{\mathrm{R}}), \tag{1}$$

where $\widetilde{I}_t$ denotes the adapted human image during training or the robot camera image at deployment, $q_t$ is the 43-D G1 state, $z_t \in \mathbb{R}^{64}$ is a SONIC motion token, and $h_t^{\mathrm{L/R}} \in \mathbb{R}^6$ are Revo2 hand

commands. Controller rollouts supply $q_t$ in the robot's state space during training. At deployment, SONIC executes motion tokens and the Revo2 driver executes hand commands. The same SONIC checkpoint serves alignment and deployment across all four tasks.

### 3.3 Action Alignment

Action alignment (Figure 3(a)) seeks accurate interaction trajectories while retaining the tracking policy's coordinated control. Starting from a controller rollout ($H$), the coarse stage corrects reference geometry ($K$), and the fine stage iteratively refines motion tokens using execution feedback ($D$).

**Motion initialization.** We obtain an initial robot motion by executing human body motion through SONIC in simulation, retaining the resulting joint trajectory $b^0_{1:T}$ and motion tokens, where $b_t \in \mathbb{R}^{29}$. We denote this initialization by $H$. Other motion references are also compatible with the framework; we compare their accuracy and motion quality in Table 3.

**Target construction.** We express motion in pelvis-centered frames with vertical axes fixed upright and horizontal axes following pelvis yaw. Human palm poses define the targets, with heights adjusted by the per-frame human–robot pelvis-height difference from the initial rollout. Targets remain fixed throughout refinement. Palm and sole target definitions appear in Appendix A.2.

#### 3.3.1 Kinematic Reference Correction ($K$)

The coarse stage uses constrained offline IK to bring the initial robot reference closer to the human targets. The optimization reduces end-effector pose error while limiting changes to the original motion and maintaining temporal continuity. We encode the corrected reference $b^{\text{ref}}_{1:T}$ using SONIC's robot motion encoder $\mathcal{E}_r$ (Luo et al., 2026):

$$z^0_{1:T} = \mathcal{E}_r(b^{\text{ref}}_{1:T}). \tag{2}$$

Token replay measures the residual tracking errors corrected in the fine stage (Appendix A.2).

#### 3.3.2 Dynamics-Aware Refinement ($D$)

The fine stage uses rollout feedback and an MLP response model $\mathcal{R}_\psi$. Paired perturbed and unperturbed rollouts train it to predict execution changes from kinematic changes (Appendix A.3).

At iteration $k$, a rollout of $z^k$ provides the controller observation history $s^k$ and end-effector residual $r^k$: position errors and, for hand targets, relative rotation vectors from executed poses to fixed targets. Holding the rollout context fixed, SONIC's control decoder $\mathcal{D}_c$ maps candidate tokens to joint commands. We predict their effect on execution as

$$\begin{aligned} \Delta u(z) &= \mathcal{D}_c(z, s^k) - \mathcal{D}_c(z^k, s^k), \\ \widehat{\Delta e}(z) &= \mathcal{R}_\psi\big(J^k \Delta u(z)\big)\,. \end{aligned} \tag{3}$$

Here, $J^k$ maps joint-command changes to kinematic end-effector changes. We optimize the tokens so that the predicted response compensates for the current residual:

$$\mathcal{L}_{\text{track}}(z) = \|\widehat{\Delta e}(z) - r^k\|^2. \tag{4}$$

Gradients pass through the fixed response model and control decoder, not the simulator. Each updated sequence is evaluated in a fresh rollout, which supplies the residual and context for the next iteration. We select a validated sequence from the optimization history for policy training. The selected output, $D_s$, may come from an earlier round, including $H$ or $H + K$. The full formulation adds temporal history, state and token conditioning, task weights, and regularization; selection criteria and implementation details appear in Appendices A.2–A.4.

### 3.4 Visual Alignment

Action alignment alone does not close the visual gap between human demonstrations and robot deployment (Table 4). We address differences in arm appearance and camera viewpoint through robot-arm rendering and training-time image augmentation (Figure 3(b)).

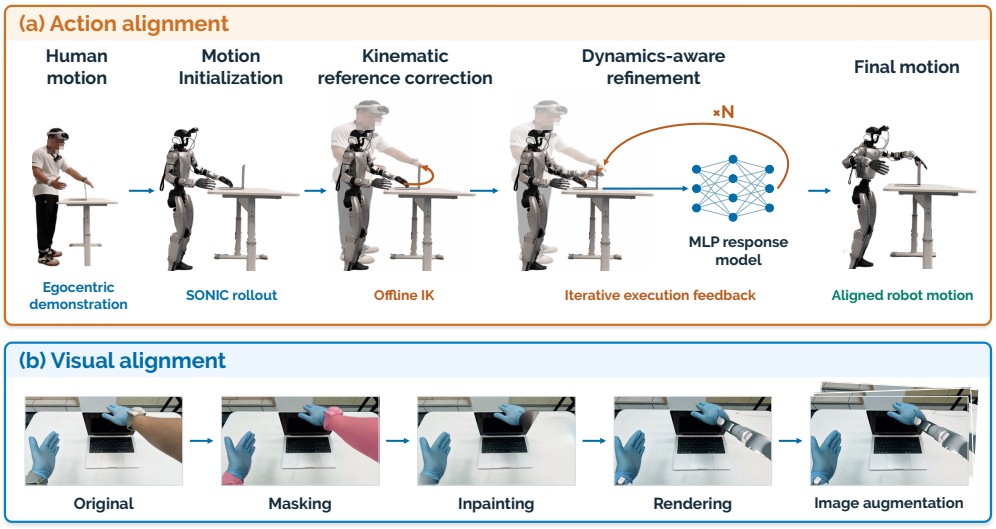



Figure 3: Action and visual alignment pipelines. (a) Human motion initializes a robot trajectory through a SONIC rollout ($H$). Offline IK corrects the reference geometry ($K$), and dynamics-aware refinement ($D$) uses an MLP response model and iterative execution feedback to produce aligned robot motion. (b) Human forearms and wrist trackers are masked and inpainted, then replaced with rendered robot arms. Image augmentation produces varied training views.

**Robot-arm rendering.** We adapt arm appearance (Chen et al., 2024; 2025). Matching blue gloves on humans during collection and robots during deployment simplify segmentation and visual alignment. We use SAM2 (Ravi et al., 2025) to segment human forearms and wrist-mounted PICO trackers, then fill the masked regions with DiffuEraser (Li et al., 2025b). We render G1 arms from aligned joint positions and adjust their translation, rotation, and uniform scale to match human wrists and forearms before compositing onto the inpainted backgrounds.

**Image augmentation.** Differences in movement amplitude and speed can alter camera viewpoints despite matching the initial camera height and angle. Random resized crops, small rotations, and photometric jitter improve robustness to these variations. At inference, the policy uses robot images directly (Appendix B.1). Table 4 evaluates rendering and augmentation separately and together.

## 4 Experiments

We evaluate skill transfer on a real humanoid and use simulation to examine the accuracy and motion quality of action alignment. We ask four questions:

**Q1:** Can human demonstrations teach whole-body loco-manipulation skills?

**Q2:** How do human and robot data compare in performance and cost?

**Q3:** How does action alignment improve accuracy?

**Q4:** How does visual alignment improve transfer?

### 4.1 Experimental Setup

We evaluate four coordinated whole-body manipulation tasks on Unitree G1, covering hand and foot interactions with posture adjustments or stepping (Figure 4).

(1) `Close Laptop`. The robot reaches above the laptop with its right hand and presses the lid shut, testing reaching accuracy and controlled downward contact until the lid is closed. (2) `Open Fridge`. The robot hooks its left hand into the door gap and pulls the door open, requiring precise hand placement and sustained contact as the robot adjusts its posture. (3) `Pull Curtain`. The robot grasps the curtain with its right hand and pulls it open by one

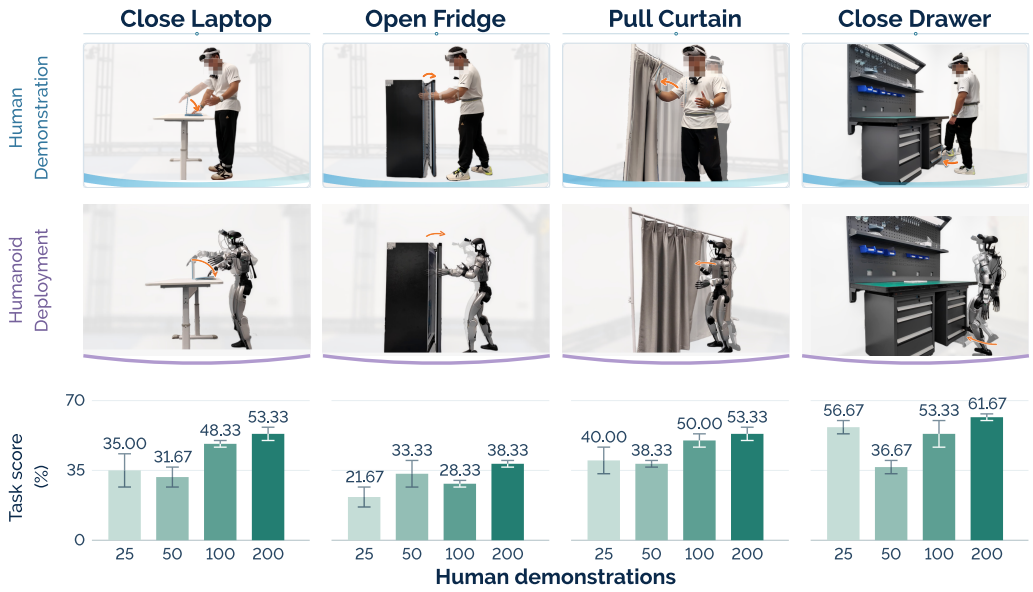



Figure 4: **Q1: Skill transfer on four real-world tasks.** Each task column shows human demonstrations (top), humanoid motion illustrations (middle), and task scores with 25, 50, 100, or 200 human demonstrations (bottom). Policies use no target-task robot demonstrations. Error bars indicate standard deviations from evaluations of the same checkpoint per condition.

meter, combining grasp maintenance with arm and body motion over an extended pulling distance. (4) `Close Drawer`. The robot lifts its right foot and pushes the open drawer fully closed with its toe, requiring accurate foot placement while maintaining balance.

For each task, we post-train a separate $\pi_{0.5}$ policy for 30,000 steps with batch size 256, using the same pretrained checkpoint and training settings (Appendix B.1). Unless otherwise specified, human demonstrations undergo the full $H + K + D_s$ action-alignment pipeline, robot-arm rendering, and image augmentation; Close Drawer omits arm rendering. Human-data policies use no target-task robot demonstrations during post-training and are deployed directly on the robot.

Each policy is evaluated over 20 real-world trials with three sequential subtasks, each scored as success or failure. Averaging over subtasks and trials gives a task score that captures partial completion (Appendix B.3). Offline endpoint-error and motion-quality comparisons use a fixed random subset of 25 training trajectories per task (Appendices C.1–C.3). Cross-task means weight tasks equally.

### 4.2 Q1: Skill Transfer from Human Demonstrations

Figure 4 evaluates transfer with 25, 50, 100, and 200 human demonstrations per task. Increasing the budget from 25 to 200 raises the four-task mean score from 38.34% to 51.67% and mean full-task success from 18.75% to 38.75% (Table 6). This scaling trend suggests that the policy learns visual generalization rather than simply overfitting to the training demonstrations.

### 4.3 Q2: Human vs. Robot Demonstrations

Figure 5 compares human and robot demonstrations on Close Laptop and Pull Curtain under the same training and evaluation settings. With 200 human demonstrations, Close Laptop reaches 50% full-task success, matching the best tested robot-data result (50 demonstrations). Pull Curtain reaches 30%, compared with the best tested robot-data result of 35% (50 demonstrations) (Tables 6–7). At fixed budgets of 200 human versus 50 robot demonstrations per task, mean full-task success is 40.00% versus 42.50%, while mean task scores are 53.33% versus 60.00%. These results show comparable full-task success with more human demonstrations.

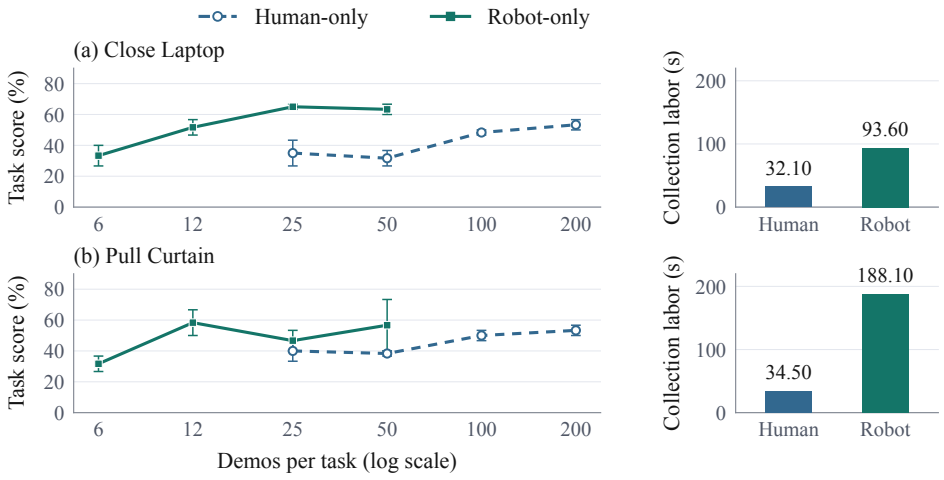



Figure 5: **Task scores and collection labor with human or robot demonstrations.** Each point represents an independently trained policy evaluated over 20 trials. Human-only results reuse Figure 4. Collection labor is measured in person-seconds per episode and includes capture, setup, calibration, troubleshooting, and resets. Error bars indicate standard deviations from evaluations of the same checkpoint per condition.

Human demonstrations require 76.36% less collection labor per episode on average (33.30 vs. 140.85 person-seconds), which can offset the need for more demonstrations. On Pull Curtain, collecting 200 human demonstrations requires 26.63% less labor than collecting 50 robot demonstrations.

### 4.4 Q3: Action Alignment for Improved Accuracy

**Why correct actions offline?** We first examine whether online IK can improve endpoint accuracy during whole-body tracking. All three tested implementations produce unstable SONIC rollouts, including falls, across the four tasks, so we do not proceed to policy evaluation with these variants. Their designs and failure sequences are reported in Appendix D.

**Kinematic reference correction and dynamics-aware refinement.** Table 1 compares $H$, $H+K$, and selected $H + K + D_s$ outputs using either unit or MLP response under the same controller. Position error covers all four tasks; orientation error covers the three palm-interaction tasks. Drawer closing uses sole position only.

Mean position error decreases from 11.21 cm at $H$ to 4.45 cm after $K$ and 3.41 cm at $H + K + D_s$; orientation error falls from 21.22° to 13.77° and 6.71°. Thus, $K$ removes much of the initial position mismatch, while $D$ further improves execution accuracy, especially in orientation. With 200 demonstrations per task, mean policy score likewise rises from 21.25% to 47.09% and 51.67%. The full pipeline improves policy scores over $H + K$ on every task. An expanded simulation study on 112 trajectories across 50 tasks shows the same stage-wise trend: position error falls from 17.03 to 5.22 to 3.83 cm (Appendix C.6, Table 12). Table 2 suggests that $D$, particularly the MLP variant, benefits both precise positioning and interaction (Close Laptop S2, Open Fridge S3, and Pull Curtain S1) and stable whole-body coordination (Close Drawer S3). The latter requires maintaining balance while pushing the drawer fully closed.

**Motion initialization.** Table 3 compares three motion initialization strategies: a SONIC rollout, GMR retargeting (Araujo et al., 2026), and whole-body IK adapted from HuMI (Nai et al., 2026). Each passes through the same subsequent $K$ stage, eight refinement rounds, and selection procedure.

SONIC rollout and retargeting achieve similar position errors (3.41 versus 3.42 cm). Retargeting gives lower orientation error (5.32° versus 6.71°), while SONIC rollout reduces extra yaw from

Table 1: **Q3: Effect of action alignment.** Top: position (cm) / orientation (°) errors in simulation (↓). Bottom: real-world task scores (%, ↑) with 200 demonstrations per task and identical visual preprocessing. Means are computed across tasks; Close Drawer uses sole position only and is excluded from orientation averages. Bold/underline mark best/second best. Unit/MLP denote response variants in $D$; see Appendices A.3, C.2, and C.5 for details.

| Alignment | Close Laptop Pos. (cm) / Ori. (°) | Open Fridge Pos. (cm) / Ori. (°) | Pull Curtain Pos. (cm) / Ori. (°) | Close Drawer Pos. (cm) | Mean Pos. (cm) / Ori. (°) |
|---|---|---|---|---|---|
| **End-effector Tracking Error (↓)** | | | | | |
| $H$ | 14.08 / 29.43 | 13.27 / 19.37 | 10.31 / 14.86 | 7.17 | 11.21 / 21.22 |
| $H + K$ | 4.22 / 13.84 | 4.24 / 14.72 | 3.13 / 12.74 | 6.22 | 4.45 / 13.77 |
| $H + K + D_s$ (Unit) | 3.18 / 9.74 | 2.89 / 5.88 | 2.19 / 5.06 | 9.18 | 4.36 / 6.89 |
| $H + K + D_s$ (MLP) | **3.04 / 9.50** | **2.76 / 5.77** | **2.02 / 4.85** | **5.80** | **3.41 / 6.71** |
| **Real-World Task Score (%, ↑)** | | | | | |
| $H$ | 0.00 ± 0.00 | 0.00 ± 0.00 | 28.33 ± 5.00 | 56.67 ± 3.33 | 21.25 |
| $H + K$ | 46.67 ± 3.33 | 31.67 ± 1.67 | 51.67 ± 1.67 | 58.33 ± 1.67 | 47.09 |
| $H + K + D_s$ (Unit) | 48.33 ± 1.67 | 33.33 ± 3.33 | **55.00** ± 5.00 | 55.00 ± 1.67 | 47.92 |
| $H + K + D_s$ (MLP) | **53.33** ± 3.33 | **38.33** ± 1.67 | 53.33 ± 3.33 | **61.67** ± 1.67 | **51.67** |

11.45° to 7.18° and improves the other motion-quality metrics. This balance of accuracy and motion quality motivates our use of SONIC rollout for motion initialization.

**Roles of $K$ and $D$.** Figure 7 supports one $K$ round followed by multiple $D$ rounds. Repeating $K$ eight times changes position error only from 4.86 to 4.77 cm. Instead, adding eight $D$ rounds after one $K$ reduces position error to 3.97 cm and orientation error from 16.46° to 8.08°. Eight $D$ rounds alone leave errors of 8.40 cm and 16.40°. Thus, $K$ provides an accurate reference, while repeated $D$ updates further reduce execution error; repeating $K$ offers little additional benefit. The 50-task experiments likewise support one $K$ round followed by multiple $D$ rounds (Figure 8).

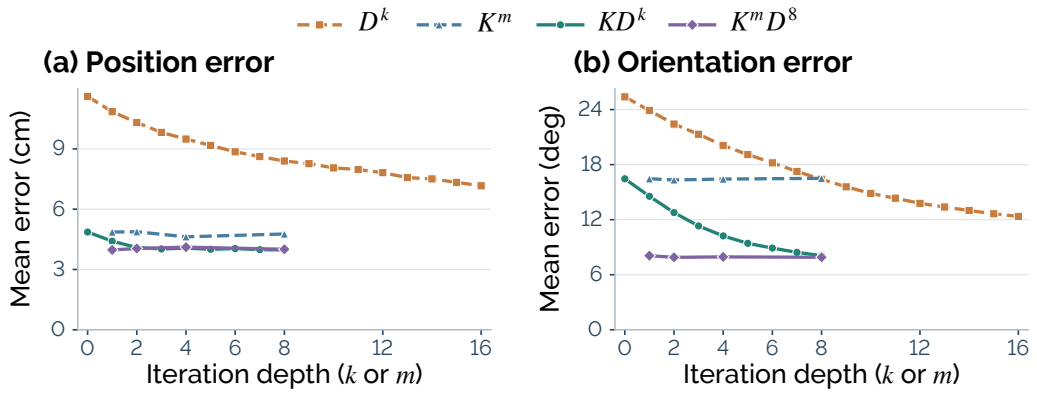



Figure 7: **Complementarity and depth of $K$ and $D$.** Task-macro averages of raw stage outputs for $D^k$, repeated $K^m$, $KD^k$, and $K^mD^8$. Position averages four tasks; orientation averages the three hand tasks. Protocol and processing costs appear in Appendix C.4.

**Learning the execution response.** Compared with the unit-response variant, the MLP variant reduces endpoint errors on all four tasks, most notably drawer position error (9.18 to 5.80 cm; Table 1). It improves real-world task scores by 5.00, 5.00, and 6.67 percentage points on Close Laptop, Open Fridge, and Close Drawer, respectively. These results suggest that learned execution responses are particularly beneficial for tasks requiring precise interactions or more demanding whole-body coordination. On 50 tasks, raw $KD^8$ errors also improve from 4.25 to 4.10 cm and 9.39° to 9.21° (Table 13; training details in Appendix A.3).

Table 2: **Stage-wise and full-task success (%).** S1–S3 are sequential; S3 denotes full-task success. Task-specific stages are defined in Appendix B.3. Bold marks the best result per column.

| Alignment | Close Laptop | | | Open Fridge | | | Pull Curtain | | | Close Drawer | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 |
| $H$ | 0 | 0 | 0 | 0 | 0 | 0 | 35 | 35 | 15 | **75** | **75** | 20 |
| $H + K$ | **65** | 45 | 30 | **50** | **35** | 10 | 65 | **65** | 25 | **75** | 70 | 30 |
| $H + K + D_s$ (Unit) | 60 | 45 | 40 | 40 | **35** | 25 | **75** | 60 | **30** | 70 | 70 | 25 |
| $H + K + D_s$ (MLP) | 60 | **50** | **50** | 45 | **35** | **35** | 70 | 60 | **30** | **75** | 70 | **40** |

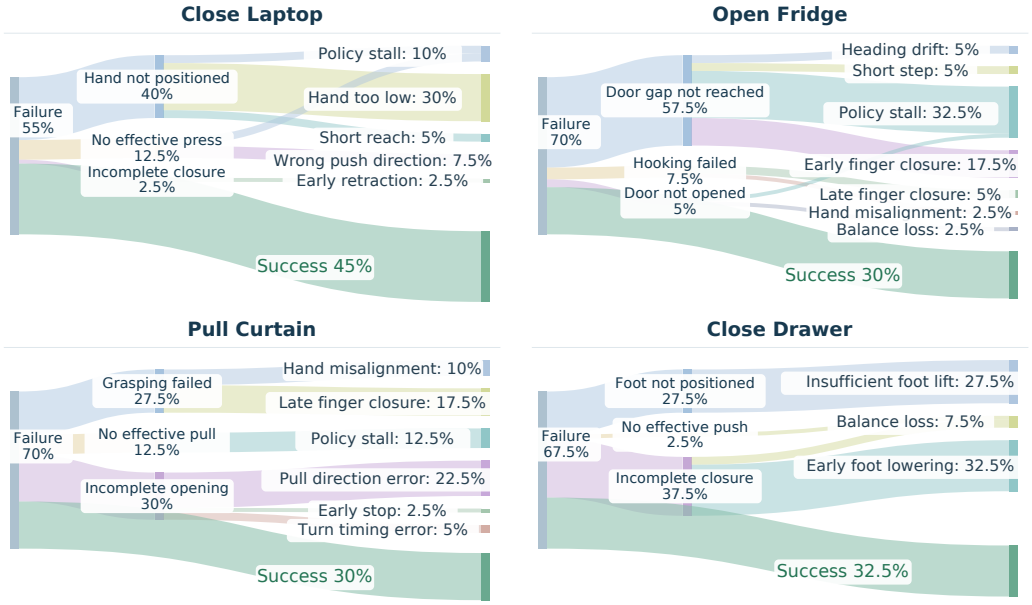



Figure 6: **Success and failure analysis of $D$ (Unit) and $D$ (MLP) (%).** 40 trials per task (20 per variant). Failures are grouped by stage and observed failure mode; shared modes merge. Detailed descriptions appear in Appendix B.4.

Table 3: **Effect of motion initialization.** Task-averaged endpoint error at $D_s$ and whole-body motion quality. Pelvis drop measures maximum height loss; tilt, maximum base inclination; joint step, maximum frame-to-frame joint-angle change; extra yaw, heading deviation from the human reference. All variants share $K$, eight $D$ rounds, and selection. Drawer orientation is excluded. Lower is better; bold/underline mark best/second best. Appendix C.3 defines the metrics; Tables 8 and 9 report results for each task and initialization.

| | **Endpoint Error** | | **Whole-Body Motion Quality** | | | |
|---|---|---|---|---|---|---|
| Motion initialization | Pos. (cm) | Ori. (°) | Pelvis drop (cm) | Tilt (°) | Joint step (°) | Extra yaw (°) |
| SONIC Rollout | **3.41** | 6.71 | **3.28** | **11.14** | **5.94** | **7.18** |
| Retargeting | <u>3.42</u> | **5.32** | <u>3.30</u> | <u>11.87</u> | <u>6.51</u> | <u>11.45</u> |
| Whole-Body IK | 3.82 | <u>6.35</u> | 5.25 | 12.77 | 7.75 | 23.67 |

### 4.5 Q4: Visual Alignment for Easier Transfer

We evaluate robot-arm rendering and image augmentation separately and together, using 200 human demonstrations per task with identical action labels and training settings (Table 4). The baseline uses human images with standard model preprocessing only.

Table 4: **Q4: Visual-alignment ablation.** Task scores with 200 human demonstrations per task. Means average the available tasks; robot-arm rendering is not evaluated for foot-operated Close Drawer. Bold/underline mark best/second best; shading indicates the full pipeline. Stage-wise results appear in Appendix B.6.

| **Visual alignment** | | **Task score (%)** | | | | **Mean** |
|---|---|---|---|---|---|---|
| Aug. | Render. | Close Laptop | Open Fridge | Pull Curtain | Close Drawer | |
| × | × | 31.67 ± 5.00 | 16.67 ± 3.33 | 35.00 ± 1.67 | <u>50.00</u> ± 3.33 | 33.34 |
| × | ✓ | 51.67 ± 1.67 | 23.33 ± 3.33 | <u>41.67</u> ± 8.33 | — | 38.89 |
| ✓ | × | **61.67** ± 8.33 | <u>33.33</u> ± 6.67 | 33.33 ± 6.67 | **61.67** ± 1.67 | <u>47.50</u> |
| ✓ | ✓ | <u>53.33</u> ± 3.33 | **38.33** ± 1.67 | **53.33** ± 3.33 | — | **48.33** |

Across the three arm tasks, rendering, augmentation, and their combination achieve mean scores of 38.89%, 42.78%, and 48.33%, respectively, with unchanged action labels. Combining them gives the best results on Open Fridge (38.33%) and Pull Curtain (53.33%), while augmentation alone is best on Close Laptop (61.67%). On Pull Curtain, combining both components improves over rendering alone by 11.66 percentage points and augmentation alone by 20.00 points. Thus, their benefits are complementary but task-dependent.

## 5 Conclusion

EgoHumanoid-V2 transfers coordinated whole-body loco-manipulation skills from egocentric demonstrations through coarse-to-fine action alignment and visual adaptation. With 200 human demonstrations per task, VLA policies achieve a mean score of 51.67% across four real-world tasks without target-task robot demonstrations, demonstrating zero-shot skill transfer on the tested tasks. The results support human motion as direct supervision for coordinated humanoid skills.

**Limitations and future work.** Alignment can still produce unstable motions, and IK and repeated rollouts limit throughput. Stronger stability constraints and faster optimization remain directions for improving reliability and processing efficiency.

## AI USE STATEMENT

Generative AI tools assisted with literature discovery and synthesis, method and experimental-design feedback, manuscript organization, drafting and editing, reference formatting, figure development, and software development. The authors reviewed this work and take responsibility for the final text, claims, citations, code, and artifacts.

## REPRODUCIBILITY STATEMENT

Sections 3 and 4 describe the learning interface and controlled comparisons. Appendices A and B provide implementation and policy evaluation details. Appendices C and D report supplementary action-alignment experiments and online-IK diagnostics.

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The appendix covers implementation (Appendix A), policy training and evaluation (Appendix B), supplementary action-alignment results (Appendix C), and online-IK diagnostics (Appendix D).

# A Implementation Details

## A.1 Data Preparation and Validation

GoPro RGB, PICO whole-body tracking, and hand-pose signals form synchronized 50 Hz image–trajectory pairs. Signal validation and heading calibration precede $H$, followed by $K$ and $D$. Accepted replays provide states and actions paired with aligned images for VLA post-training.

Each training sample uses the 43-dimensional state and the 76-dimensional action interface in Eq. 1.

Validation checks PICO–GoPro timing consistency, video/frame parity, endpoint tracking, joint increments, base tilt, and pelvis displacement. Without an external pulse, timing consistency measures repeatability rather than absolute synchronization error. These checks assess training-data validity, not task success.

## A.2 Action Alignment Details

**Coordinate frames and target registration.** A world-space point $p_t^W$ is expressed in the local pelvis frame as

$$p_t^L = R_z(\psi_t)^\top (p_t^W - p_{\text{pelvis},t}^W), \tag{5}$$

where $\psi_t$ is pelvis yaw. The frame has a vertical $z$ axis and removes world translation and heading, but not body tilt. Human palm targets retain local horizontal coordinates and orientation. Their vertical coordinate is $y_{t,z} = p_{t,z}^{\mathrm{H},L} + h_t^{\mathrm{H}} - h_t^{\mathrm{R},0}$, where $h_t^{\mathrm{H}}$ and $h_t^{\mathrm{R},0}$ are pelvis heights above the floor in the human recording and initial robot generation. Registration remains fixed across rounds. Human-conditioned controller execution supplies the initial joint reference; independent token replay measures tracking error against the fixed human palm targets.

Foot-position targets use the midpoint of human ankle and foot keypoints as a sole-center proxy, not a measured contact point. We estimate an episode-level floor offset from the human recording and apply the same pelvis-height registration. Foot-only tasks omit the orientation objective.

**Kinematic reference correction ($K$).** Both $K$ and $D$ target bilateral hand poses by default; foot-only tasks such as Close Drawer target only foot positions. For hand tasks, $K$ optimizes only the arms and waist. Both IK solvers constrain deviations from the initial reference.

**Dynamics-aware refinement ($D$).** Joint-preservation weights for the legs, waist, and arms are $(20, 20, 0.1)$ for hand tasks and $(0.1, 20, 20)$ for foot-only tasks. These soft penalties regularize proposed corrections without freezing body parts during execution.

Using the response prediction in Eq. 3, we weight the endpoint residual by the task metric $Q_\tau$. The context $c^k$ and observation history $s^k$ remain fixed during the local solve. We optimize

$$z^{k+1} = \arg\min_{z \in \mathcal{N}(z^k)} \|\widehat{\Delta e}(z) - r^k\|_{Q_\tau}^2 + \mathcal{L}_{\text{reg}}(z, z^k). \tag{6}$$

For each frame, let $\delta_t = z_t - z_t^k$ and $\Delta u_t = u_t(z) - u_t(z^k)$ denote the token and decoded joint-command changes. The neighborhood is

$$\mathcal{N}(z^k) = \{z^k + \delta : \ \|\delta_t\|_\infty \le \epsilon_\infty, \ \|\delta_t\|_2 \le \epsilon_2 \text{ for every } t\}.$$

We parameterize $\delta_t$ with an elementwise scaled $\tanh$ and rescale any row exceeding the L2 bound. Token parameters start at zero each round.

The regularizer sums weighted joint preservation, token magnitude $\lambda_z \,\mathrm{mean}(\delta^2)$, token smoothness $\lambda_{\Delta z}\,\mathrm{mean}((D_1\delta)^2)$, joint-correction smoothness $\lambda_v\,\mathrm{mean}(v \odot (D_1\Delta u)^2)$ and $\lambda_a\,\mathrm{mean}(a \odot (D_2\Delta u)^2)$, and an IK correction prior with weight $\lambda_{\mathrm{IK}}$. Here, $D_1$ and $D_2$ are first and second frame differences at 50 Hz, without division by the frame interval. The multipliers $(v, a)$ are $(8, 32)$ for hips and shoulders, $(3, 8)$ for knees and elbows, $(16, 64)$ for the waist, and $(1, 1)$ for ankles and

wrists. Joint preservation uses the weights above and averages over frames and all 29 joints; the IK prior averages squared error to the corrected joint target over frames and active joints (arms for hand tasks, legs for foot tasks).

| Setting | Regular $D$ | Conservative $D$ |
|---|---|---|
| Token bounds $(\epsilon_\infty, \epsilon_2)$ | $(0.015, 0.05)$ | $(0.005, 0.02)$ |
| Token magnitude $\lambda_z$ | 0.002 | 0.005 |
| Token smoothness $\lambda_{\Delta z}$ | 0.002 | 0.008 |
| Joint first-difference weight $\lambda_v$ | 0.002 | 0.006 |
| Joint second-difference weight $\lambda_a$ | 0.004 | 0.012 |
| IK prior $\lambda_{\text{IK}}$ | 0.1 | 0.1 |
| Adam learning rate | 0.01 | 0.008 |

Each round uses 400 Adam steps without early stopping through the frozen decoder and response model. The default schedule is one $K$ round, seven regular $D$ rounds, and one conservative $D$ round. Before replay, token corrections are filtered with a quadratic Savitzky–Golay filter (15/21 frames for regular/conservative rounds), clipped to the same bounds, and limited to per-element frame differences of 0.005/0.002; the first 47 correction frames remain zero. The initial $H$ token prefix is retained for 10 frames and blended into the candidate over the next 50 frames. Each resulting sequence is evaluated by a fresh rollout; historical candidate selection follows separately.

### A.3 Response Model and Training

**Inputs and temporal context.** The response predictor $\mathcal{R}_\psi$ in Eq. 3 includes causal history and conditions on fixed baseline features $c^k$, omitted from the main-text notation. This history is distinct from initialization $H$. Position and rotation changes for both palms and soles form a 24-D input. The current input and five exponentially filtered copies (time constants: 0.02, 0.1, 0.5, 2, and 5 s) form a 144-D causal history, conditioned on 93 robot-state features and the 64-D reference token.

**Response training data.** We construct response supervision from the G1 motion trajectories in BONES-SEED[1]. We replay reference and perturbed token sequences from identical simulator states and controller histories at 50 Hz. Perturbations use smooth random changes and token-refinement directions. Inputs are Jacobian-based endpoint changes between commands decoded under the same baseline observation; targets are executed endpoint-pose differences one policy step later. Training uses 250 motions from 68 capture dates (345,080 frame–perturbation pairs); validation uses 13 motions from seven separate dates (20,420 pairs).

**Architecture and optimization.** A frozen linear response and an MLP residual jointly predict the 24-dimensional endpoint response. The MLP has two hidden layers of width 64 with SiLU activations. Subtracting its output at zero history enforces zero response when the intervention history is zero. We normalize inputs and targets using training-set statistics and minimize normalized mean squared error with a squared residual penalty of weight 1. Random perturbations receive 75% of the training weight; the two refinement-derived direction families share the remaining 25%. Within each family, capture dates and source motions are balanced. We use AdamW with learning rate $3 \times 10^{-5}$, weight decay $10^{-4}$, batch size 512, and gradient clipping at norm 5. Training runs for at most 150 epochs, retaining the checkpoint with the lowest validation loss. During token optimization, model weights and baseline features remain fixed. Gradients traverse the candidate-dependent history, not the simulator.

**Endpoint configurations.** For palm tasks, we blend the learned response with the unit response using a learned-response weight of 0.15. For foot-only tasks, we use the learned position response without an orientation objective. Configurations are chosen through task-level development rollouts and fixed across episodes and refinement rounds.

[1] https://huggingface.co/datasets/bones-studio/seed

### A.4 Motion Selection and Validation

Historical candidates are evaluated through simulation rollouts and ranked using task-dependent tracking and motion metrics. Selection may retain $H$ or $H + K$, while demonstrations are rejected if no candidate satisfies the pelvis-displacement criterion. As summarized in Table 11, $D$ candidates are selected for 97.1% of inputs on average, with 1.84% rejected. We retain the top 200 accepted demonstrations per task, ranked by the selection metrics, for policy training. Diagnostic selection and aggregation follow Appendix C.

Before VLA training, hand commands for Open Fridge and Pull Curtain are binarized to simplify learning, and motion trajectories are lightly filtered. For Pull Curtain, trailing segments containing incidental demonstrator movements are trimmed during data preparation. All tasks reuse the same low-level checkpoint without target-task training or fine-tuning. This does not imply that its pretraining excluded similar motions.

## B Policy Training and Evaluation

### B.1 Post-Training Configuration

We initialize $\pi_{0.5}$ from `pi05_base` and post-train in bfloat16 for 30,000 steps with global batch size 256. We use AdamW, gradient clipping at 1.0, and weight decay $10^{-10}$. A cosine schedule has 1,000 warmup steps, a peak learning rate of $2.5 \times 10^{-5}$, and a terminal rate of $2.5 \times 10^{-6}$. These settings are shared across controlled comparisons.

**Visual preprocessing.** Robot-arm rendering follows Sec. 3.4 offline. Post-training adds random resized crops, in-plane rotation, and photometric jitter. Deployment uses robot images with standard preprocessing only.

### B.2 Training Data and Controlled Comparisons

Human data use $H + K + D_s$, robot-arm rendering, and image augmentation unless ablated; foot-operated Close Drawer omits arm rendering. Q1 uses 25, 50, 100, or 200 human demonstrations per task; action and visual ablations use 200. Q2 trains separate human-data and robot-data policies at the budgets in Sec. 4.3. All conditions share the policy backbone, action interface, and evaluation protocol.

In Table 1, $H$ and $H+K$ supply stage-specific action labels; the full variant uses validated historical candidates (Appendix A.4). Visual preprocessing and training settings are identical.

### B.3 Task Scoring Criteria

Each task comprises three ordered stages, S1–S3. Full-task success requires completing all three stages. Evaluation stops at the first failed stage; that stage and all subsequent stages receive zero. We also report partial completion through the task score: each trial receives $0$, $1/3$, $2/3$, or $1$, averaged over the 20 attempted trials per condition:

$$\text{Task score } (\%) = \frac{100}{20} \sum_{i=1}^{20} \left( \frac{1}{3} \sum_{j=1}^{3} s_{ij} \right), \qquad s_{ij} \in \{0, 1\}. \tag{7}$$

Table 2 reports completion rates for each stage; the S3 rate is the full-task success rate.

**Close Laptop.** The robot presses the laptop lid closed with its right hand.

- **S1** The right hand reaches a suitable pressing position above the laptop lid.
- **S2** The right hand applies an effective downward press to the lid.
- **S3** The lid is fully closed.

**Open Fridge.** The robot opens the refrigerator door with its left hand.

- **S1** The left hand reaches the vicinity of the door gap.
- **S2** The left hand hooks into the gap.
- **S3** The refrigerator door opens.

**Pull Curtain.** The robot pulls the curtain open by one meter with its right hand.

- **S1** The right hand grasps the curtain.
- **S2** The right hand pulls and moves the curtain.
- **S3** The curtain is pulled open by one meter.

**Close Drawer.** The robot pushes the lowest drawer closed with its right foot.

- **S1** The right foot lifts, with its toe positioned in front of the drawer's outer face.
- **S2** The right foot pushes the drawer inward.
- **S3** The right foot pushes the drawer fully closed.

### B.4 Failure-Case Analysis

Figure 6 pools 40 trials per task: 20 each from $D$ (Unit) and $D$ (MLP). The figure reports percentages of all 40 trials; the counts below refer to this same pool. Each failed trial is assigned to its first unsuccessful stage and one primary observed failure mode. Stage definitions follow Appendix B.3.

**Close Laptop.** 18 trials succeed and 22 fail. Of 16 S1 failures, the policy stalls during approach (2), the hand is too low and collides with the laptop (12), or forward reach is insufficient (2). Of 5 S2 failures, the policy stalls without pressing (2), or the push direction causes the lid to reopen or jam (3). The single S3 failure is early hand retraction before full closure. The two stall branches merge as *Policy stall* (4 trials, 10%). Stalls may reflect out-of-distribution visual inputs or a visual gap; early retraction may reflect insufficient visual discrimination of partial versus complete closure.

**Open Fridge.** 12 trials succeed and 28 fail. Of 23 S1 failures, stepping-induced sway causes heading drift (2), the step is too small (2), the policy stalls without an effective forward reach (12), or the fingers close too early (7). Of 3 S2 failures, late finger closure (2) or inaccurate hand positioning (1) prevents hook engagement. The 2 S3 failures are policy stalling without an effective opening action (1) and loss of balance while stepping backward (1). The S1 and S3 stalls merge as *Policy stall* (13 trials, 32.5%).

**Pull Curtain.** 12 trials succeed and 28 fail. Of 11 S1 failures, the hand is misaligned (4), or the fingers close too late (7). All 5 S2 failures are policy stalls without an effective pull. Of 12 S3 failures, the arm's pulling direction is substantially misaligned with the curtain's sliding direction (9), the robot stops too early (1), or turning stops too early or too late, leaving an incorrect travel heading (2).

**Close Drawer.** 13 trials succeed and 27 fail. All 11 S1 failures involve unsuitable foot positioning, primarily insufficient lift. The single S2 failure is loss of balance before an effective push. Of 15 S3 failures, the robot stumbles before full closure (2), or lowers its foot too early (13). The S2 and S3 balance failures merge as *Balance loss* (3 trials, 7.5%). Insufficient foot lift (11) and early foot lowering (13) remain separate causes.

### B.5 Result Aggregation

Task scores measure partial completion, not full-task success rates. Means weight tasks equally over the available scores; visual-ablation rows with arm rendering exclude Close Drawer. Tables 1 and 4 report task scores as mean $\pm$ standard deviation. Within each task, all models are trained with the same random seed. Reported standard deviations reflect evaluations of the same checkpoint for each condition. Policy trials are separate from endpoint-error evaluations. Object/robot pose

perturbations (position and orientation) test within-task execution in the same scenes, not unseen-task generalization.

### B.6 Stage-wise Results for Visual Alignment

Table 5 provides stage-wise completion and full-task success for the visual-alignment ablation in Table 4, using the criteria in Appendix B.3. Close Drawer uses image augmentation without arm rendering, so its corresponding full-pipeline results appear in the augmentation-only row.

Table 5: **Stage-wise success for visual alignment (%).** S3 denotes full-task success. Arm rendering is not evaluated for Close Drawer. Bold marks the best result per column.

| Visual alignment | | Close Laptop | | | Open Fridge | | | Pull Curtain | | | Close Drawer | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Aug. | Render. | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 |
| × | × | 40 | 30 | 25 | 25 | 15 | 10 | 50 | 45 | 10 | 65 | 60 | 25 |
| × | ✓ | **75** | 40 | 40 | 35 | 20 | 15 | 55 | 50 | 20 | — | — | — |
| ✓ | × | 70 | **60** | **55** | **45** | 30 | 25 | 40 | 40 | 20 | **75** | **70** | **40** |
| ✓ | ✓ | 60 | 50 | 50 | **45** | **35** | **35** | **70** | **60** | **30** | — | — | — |

### B.7 Stage-wise Results across Demonstration Budgets

Tables 6 and 7 report stage-wise completion for the human- and robot-demonstration budgets in Figures 4 and 5. S3 denotes full-task success.

Table 6: **Stage-wise success with human demonstrations (%).** S3 denotes full-task success.

| Demonstrations | Close Laptop | | | Open Fridge | | | Pull Curtain | | | Close Drawer | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 | S1 | S2 | S3 |
| 25 | 45 | 40 | 20 | 30 | 20 | 15 | 55 | 50 | 15 | 75 | 70 | 25 |
| 50 | 50 | 25 | 20 | 45 | 30 | 25 | 45 | 45 | 25 | 50 | 40 | 20 |
| 100 | 70 | 40 | 35 | 35 | 25 | 25 | 65 | 60 | 25 | 65 | 65 | 30 |
| 200 | 60 | 50 | 50 | 45 | 35 | 35 | 70 | 60 | 30 | 75 | 70 | 40 |

Table 7: **Stage-wise success with robot demonstrations (%).** S3 denotes full-task success.

| Demonstrations | Close Laptop | | | Pull Curtain | | |
|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | S1 | S2 | S3 |
| 6 | 55 | 35 | 10 | 45 | 40 | 10 |
| 12 | 75 | 50 | 30 | 80 | 70 | 25 |
| 25 | 85 | 65 | 45 | 65 | 60 | 15 |
| 50 | 80 | 60 | 50 | 70 | 65 | 35 |

## C Supplementary Action-Alignment Experiments

We supplement Sec. 4.4 with four-task protocols and diagnostics (Appendices C.1–C.5) and a separate 50-task simulation study (Appendix C.6).

### C.1 Shared Protocol and Metrics

Four-task comparisons use the same fixed random sample of 25 training trajectories per task (Sec. 4.1), with matched targets, registration, controller checkpoint, reset states, and evaluation windows. They measure replay accuracy separately from autonomous policy performance.

Table 8: **Task-wise endpoint error for motion initialization.** Position/orientation errors at $D_s$ are in cm/°, averaged over 25 episodes per task. Drawer orientation is not evaluated. Lower is better; bold and underline denote the best and second-best values for each metric.

| Task | SONIC Rollout | Retargeting | Whole-Body IK |
|---|---|---|---|
| Close Laptop | 3.04 / 9.50 | **2.67** / <u>6.33</u> | <u>2.75</u> / **6.19** |
| Open Fridge | **2.76** / <u>5.77</u> | <u>3.15</u> / **5.74** | 3.68 / 7.25 |
| Pull Curtain | **2.02** / <u>4.85</u> | <u>2.05</u> / **3.88** | 2.73 / 5.61 |
| Close Drawer | **5.80** / — | <u>5.80</u> / — | 6.11 / — |

Position error is measured in centimeters and orientation error in degrees. Errors are averaged over frames and the relevant endpoints within each trajectory, then over trajectories. Cross-task means weight tasks equally. Position includes both palms for the three hand tasks and both soles for Close Drawer. Orientation means include only the three hand tasks. Missing measurements are not assigned numerical errors or replaced with a more favorable round. Reported metrics use two decimal places; averaging and ranking precede rounding.

### C.2 Stage Comparison and Historical Selection

This subsection describes the offline error comparison in Table 1; training-data selection is described in Appendix A.4. The three hand tasks rank historical candidates using

$$S = 0.75\,\frac{e_{\text{pos}}}{5\,\text{cm}} + 0.25\,\frac{e_{\text{ori}}}{15^\circ}. \tag{8}$$

Here, $e_{\text{pos}}$ and $e_{\text{ori}}$ are mean bilateral palm errors from the same candidate. Close Drawer is evaluated using mean bilateral sole-position error. Historical selection considers $H$, $H + K$, and all recorded $D$ rounds, and may retain an earlier candidate.

The endpoint-error block directly uses measurements from existing candidate rollouts. No training-data acceptance filter is applied.

### C.3 Motion Initialization and Whole-Body Motion Quality

We compare SONIC Rollout with GMR retargeting and HuMI-style whole-body IK (Nai et al., 2026), adapted to our 29-joint model. All references undergo encoding and replay through the same SONIC checkpoint, then one $K$ and eight $D$ rounds. This isolates initialization, not the cited methods' full pipelines.

For each initializer, $D_s$ is selected from $D^1$–$D^8$ after $K$ based on endpoint tracking errors: bilateral palm position and orientation for hand tasks, and bilateral sole position for Close Drawer. No training-data acceptance filter is applied. Table 3 reports task-averaged endpoint and motion-quality results; Tables 8 and 9 give the task-wise breakdowns.

**Motion-quality metrics.** Tables 3 and 9 report four complementary diagnostics. **Pelvis drop** is the maximum decrease in pelvis height from its initial value, in cm. **Tilt** is the maximum angle between the base's vertical axis and world vertical. **Joint step** is the largest absolute change in any of the 29 body-joint angles between consecutive replay frames, measuring abrupt joint motion rather than walking step length. **Extra yaw** compares robot and human heading changes. We unwrap each yaw trajectory and subtract its initial angle, yielding $\psi^r$ and $\psi^h$. The metric is

$$e_{\text{yaw}} = \max\left\{\left|\min_t \psi_t^r - \min_t \psi_t^h\right|,\ \left|\max_t \psi_t^r - \max_t \psi_t^h\right|,\ \left|\psi_T^r - \psi_T^h\right|\right\}. \tag{9}$$

This captures excessive or insufficient rotation at signed turning extrema and final heading. Angular metrics use degrees. Diagnostics are computed per trajectory, averaged within tasks, then equally across tasks.

Table 9: **Task-wise whole-body motion quality for motion initialization.** Metrics are averaged over the same fixed random 25-trajectory cohort per task as Table 3, using the definitions in Appendix C.3. Lower is better. Bold and underline mark the best and second-best values per task.

| Task | Method | Pelvis drop (cm) | Tilt (°) | Joint step (°) | Extra yaw (°) |
|---|---|---|---|---|---|
| Close Laptop | SONIC Rollout | **0.28** | **9.62** | **2.87** | <u>3.94</u> |
| | Retargeting | <u>0.50</u> | <u>10.86</u> | <u>3.95</u> | **3.67** |
| | Whole-Body IK | 2.06 | 11.32 | 6.07 | 22.60 |
| Open Fridge | SONIC Rollout | <u>3.51</u> | **10.39** | **7.79** | **7.39** |
| | Retargeting | **3.35** | <u>11.62</u> | <u>8.28</u> | <u>13.96</u> |
| | Whole-Body IK | 8.08 | 13.04 | 8.42 | 16.70 |
| Pull Curtain | SONIC Rollout | **1.94** | **5.32** | <u>5.31</u> | <u>6.55</u> |
| | Retargeting | <u>2.48</u> | <u>6.16</u> | **5.26** | **5.66** |
| | Whole-Body IK | 2.92 | 7.14 | 5.92 | 7.24 |
| Close Drawer | SONIC Rollout | <u>7.40</u> | <u>19.22</u> | **7.78** | **10.84** |
| | Retargeting | **6.87** | **18.83** | <u>8.58</u> | <u>22.51</u> |
| | Whole-Body IK | 7.95 | 19.57 | 10.60 | 48.16 |

### C.4 Refinement Depth and Processing Cost

In Figure 7, $K^m D^k$ denotes $m$ kinematic reference correction rounds followed by $k$ dynamics-aware refinement rounds, starting from the SONIC Rollout reference. We compare $D^8$ with $KD^8$ and extend the $D$-only branch to 16 rounds. Repeated-$K$ comparisons use $m \in \{1, 2, 4, 8\}$, both alone and followed by eight $D$ rounds. The $KD^k$ curve reports the raw output at each depth $k = 0, \ldots, 8$. These curves assess refinement depth without historical selection.

Table 10 pairs processing time with the endpoint errors in Figure 7. Elapsed times span job start to capture completion, including preparation, optimization, and replay. Reused $H/K$ prefixes are charged at their recorded cost. Queueing, discarded failed attempts, and subsequent candidate selection and validation are excluded. Episode times are normalized by frame count, averaged within tasks, then averaged equally across tasks.

The timing runs use workstations with Ryzen 9 9950X3D CPUs and RTX 4090 GPUs (48 GB VRAM). Each machine is configured for three episode workers; $K/D$ rounds within an episode remain sequential. $K^1 D^8$ costs 1.49 s/frame; increasing $K$ depth to eight raises this to 3.38 s/frame with similar endpoint error.

Table 10 reports $K/D$ processing time measured within each episode worker process and normalized by frame count. Table 11 further reports total elapsed time for each task's complete data processing pipeline on four workstations with 12 parallel episode workers (three per workstation), including task-specific data preparation, action and visual alignment, validation, and candidate selection (Appendix A.4).

### C.5 Response-Model Ablation

Table 1 compares the learned response (Appendix A.3) with a unit response that retains the Jacobian and token optimizer but omits learned corrections. Both share $H/K$ inputs, targets, refinement settings, and eight $D$ rounds; endpoint configurations are fixed across episodes. Position and orientation errors come from the same historically selected candidate, unlike the raw outputs in Figure 7 (Appendix C.4). The real-world block of Table 1 additionally compares policies trained with unit- and MLP-response alignment at 200 human demonstrations per task.

### C.6 Action Alignment across 50 Tasks

**Scope and protocol.** Stage, refinement-depth, and response comparisons share 112 paired trajectories across 50 tasks, separate from the four-task cohort, with equal task weights. This simulation

Table 10: **Processing cost and endpoint accuracy.** Representative configurations from Figure 7. Time includes preparation, optimization, and replay, with reused $H/K$ costs included. It excludes queueing, discarded failed attempts, and subsequent selection and validation. Values are equal-task means; orientation averages the three hand tasks. Errors use raw outputs at each depth (Figure 7).

| Configuration | Time (s/frame) | Pos. (cm) | Ori. (°) |
|---|---|---|---|
| $K^1$ | 0.65 | 4.86 | 16.46 |
| $K^8$ | 2.55 | 4.77 | 16.51 |
| $D^8$ | 1.27 | 8.40 | 16.40 |
| $D^{16}$ | 2.11 | 7.16 | 12.36 |
| $K^2D^8$ | 1.80 | 4.04 | 7.90 |
| $K^4D^8$ | 2.33 | 4.11 | 7.94 |
| $K^8D^8$ | 3.38 | 4.01 | 7.90 |
| $K^1D^8$ (Ours) | 1.49 | 3.97 | 8.08 |

Table 11: **Data selection and processing cost.** Demos counts all input demonstrations. Reject, $H$, $K$, and $D$ report mutually exclusive outcomes as percentages of all inputs. Mean $D$ round averages the selected refinement-round index only over demonstrations assigned to $D$. Complementing the per-worker $K/D$ timings in Table 10, time covers the complete data processing pipeline: task-specific preparation (Appendix A.4), action and visual alignment, validation, and candidate selection, using 12 worker processes across four workstations. Average gives equal-task means.

| Task | Demos | Reject (%) | $H$ (%) | $K$ (%) | $D$ (%) | Mean $D$ round | Time (h) |
|---|---|---|---|---|---|---|---|
| Close Laptop | 312 | 0.96 | 0.00 | 0.00 | 99.0 | 6.8 | 0.67 |
| Open Fridge | 228 | 3.07 | 0.00 | 0.00 | 96.9 | 5.2 | 0.44 |
| Pull Curtain | 251 | 0.00 | 0.00 | 0.00 | 100 | 7.1 | 0.51 |
| Close Drawer | 270 | 3.33 | 4.44 | 0.00 | 92.2 | 3.3 | 0.38 |
| Average | 265.25 | 1.84 | 1.11 | 0.00 | 97.1 | 5.60 | 0.50 |

study evaluates motion alignment, not autonomous policies. Stage comparisons use historical selection; depth and response comparisons use outputs at the specified round.

**Kinematic correction and dynamics-aware refinement.** Table 12 extends the stage-wise gains in Table 1: $K$ reduces position/orientation errors from 17.03 cm/36.06° to 5.22 cm/21.81°; refinement and historical selection further reduce them to 3.83 cm/10.85°.

Table 12: **Stage comparison on the expanded benchmark.** Task-equal endpoint errors over 112 paired trajectories across 50 tasks; lower is better. $H + K + D_s$ uses historical candidate selection.

| Stage | Pos. (cm) | Ori. (°) |
|---|---|---|
| $H$ | 17.03 | 36.06 |
| $H + K$ | 5.22 | 21.81 |
| $H + K + D_s$ | **3.83** | **10.85** |

**Refinement depth.** Figure 8 shows the raw stage outputs at matched refinement depths. Increasing $K$ depth from one to eight changes position error from 5.22 to 5.01 cm. One $K$ round followed by eight $D$ rounds reaches 4.10 cm and 9.21°, compared with 5.22 cm and 21.81° after $K$ alone. Eight $D$ rounds without $K$ yield 14.16 cm and 21.91°. As in Figure 7, repeated $K$ offers little gain, while $K$ followed by $D$ outperforms either stage alone.

**Learned execution response.** After one $K$ and eight $D$ rounds, the MLP lowers position error from 4.25 to 4.10 cm and orientation error from 9.39° to 9.21° (Table 13). These modest gains use matched-depth outputs, unlike the historical selections in Table 1.

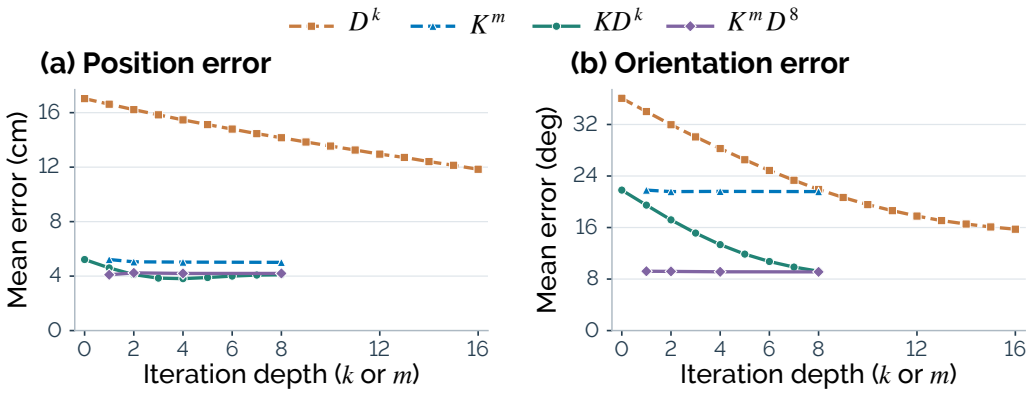



Figure 8: **Refinement depth on the expanded benchmark.** Task-equal averages of raw stage outputs for $D^k$, repeated $K^m$, $KD^k$, and $K^mD^8$ on paired trajectories.

Table 13: **Response-model comparison on the expanded benchmark.** Task-equal errors over 112 paired trajectories across 50 tasks after one $K$ and eight $D$ rounds, without historical selection; lower is better.

| Response | Pos. (cm) | Ori. (°) |
|---|---|---|
| Unit response | 4.25 | 9.39 |
| MLP response | **4.10** | **9.21** |

# D Online IK under a Whole-Body Tracking Controller

Online IK trades endpoint correction against coordinated balance. We test three SONIC command compositions on five trajectories per task: 60 runs across four tasks, separate from Appendix C. The IK objective includes position and orientation (orientation weight 0.01). Runs are neither accuracy-filtered nor retried for better outcomes.

## D.1 Three Command Compositions

For the three hand tasks, **A** assigns the arms and waist to IK while SONIC controls the legs. **B** adds SONIC's balance contribution to A. **C** restricts IK to the arms, leaving the waist and legs under SONIC control. For Close Drawer, A and B instead assign both legs to IK, without and with the balance contribution. C assigns only the right operating leg to IK; SONIC controls the remaining joints. The base and limbs remain dynamically coupled through balance and contact forces, even where SONIC commands are unchanged.

## D.2 Accuracy and Stability

Table 14 averages tracking errors over frames, both endpoints, and episodes. Tilt follows the episode-maximum definition in Appendix C.3. Fall events can recur within a run and are not failed-episode counts. Initialization failures have no formal trajectory error and are counted separately.

Video inspection finds instability in all 60 runs. All 15 drawer runs fall during first-pose preparation; the 45 hand-task runs reach their final frames without stable tracking. Variant C logs nine laptop and one fridge fall events. Variant B remains visibly unstable despite 1.50–2.43 cm hand-task position error. We therefore exclude these variants from policy training. This finding is limited to the tested SONIC implementations.

Table 14: **Online-IK stability diagnostics on five episodes per task.** Errors use measured endpoints, not IK solutions. Init. falls counts failed runs; Events counts logged falls, including startup. Dashes indicate no formal replay. Bold/underline rank values within each task from lowest to second lowest; they do not indicate that execution is stable.

| Variant | Task | Pos. (cm) | Ori. (°) | Tilt (°) | Init. falls | Events |
|---|---|---|---|---|---|---|
| A | Close Laptop | <u>4.45</u> | <u>10.43</u> | <u>16.97</u> | 0/5 | **0** |
| | Open Fridge | <u>2.33</u> | <u>15.34</u> | <u>18.97</u> | 0/5 | **0** |
| | Pull Curtain | <u>1.58</u> | <u>10.17</u> | <u>7.03</u> | 0/5 | **0** |
| | Close Drawer | — | — | — | 5/5 | — |
| B | Close Laptop | **2.43** | **8.28** | **13.30** | 0/5 | **0** |
| | Open Fridge | **1.55** | **14.45** | **17.26** | 0/5 | **0** |
| | Pull Curtain | **1.50** | **10.07** | **6.61** | 0/5 | **0** |
| | Close Drawer | — | — | — | 5/5 | — |
| C | Close Laptop | 26.50 | 40.93 | 82.75 | 0/5 | <u>9</u> |
| | Open Fridge | 10.61 | 20.55 | 34.35 | 0/5 | <u>1</u> |
| | Pull Curtain | 9.00 | 18.66 | 10.87 | 0/5 | **0** |
| | Close Drawer | — | — | — | 5/5 | — |

### D.3 Qualitative Failure Sequences

Figures 9–11 show one simulation sequence per implementation and task, chosen by peak base tilt across the five runs, including initialization. These examples illustrate instability, not its frequency.

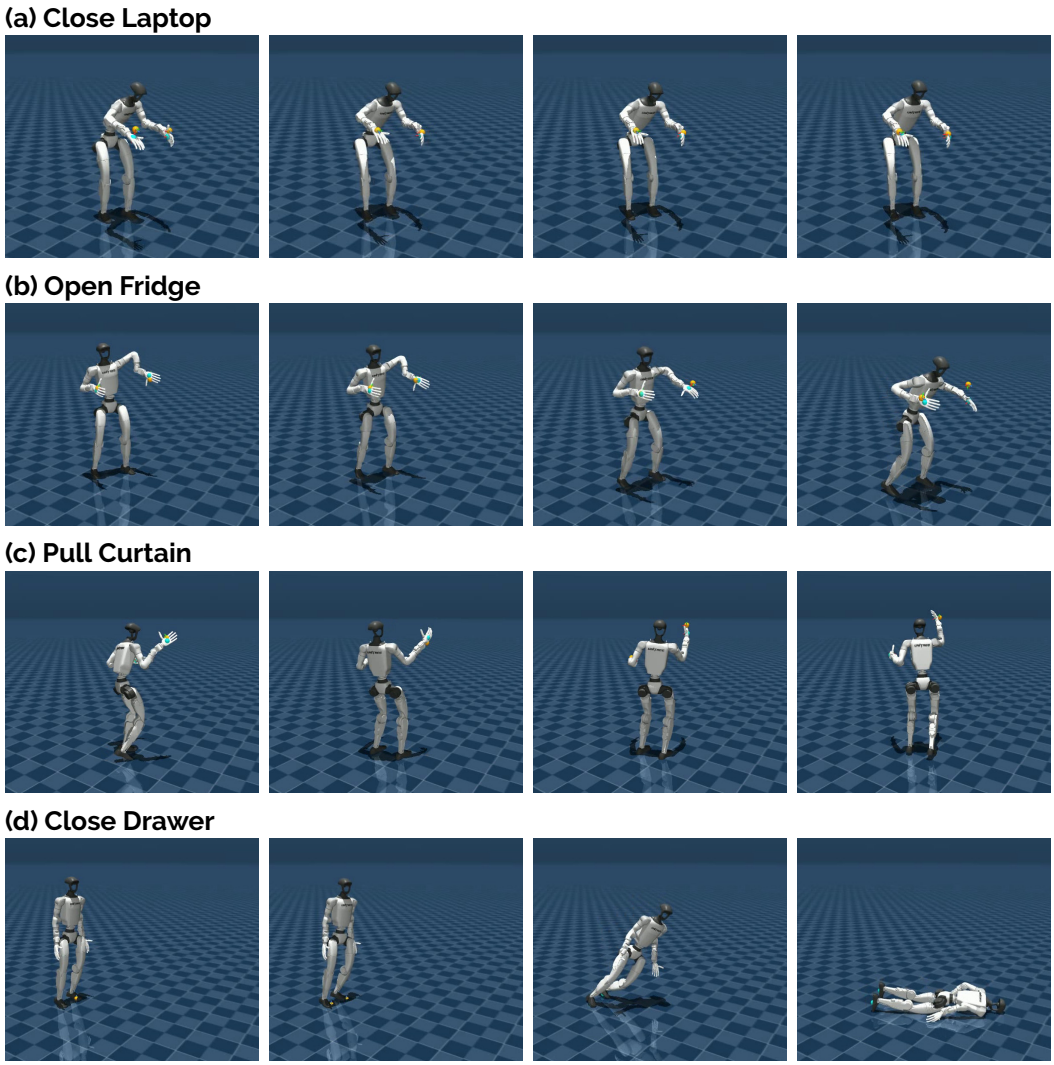



Figure 9: **Online-IK instability: variant A (IK without balance overlay).** Each row follows one continuous segment from left to right, selected by peak base tilt as described in Appendix D.3. A fixed crop within each sequence preserves relative motion and keeps robot scale comparable across tasks. Close Drawer ends in an initialization fall. The visualizations show recorded robot states without task objects; they do not depict autonomous policy evaluation.

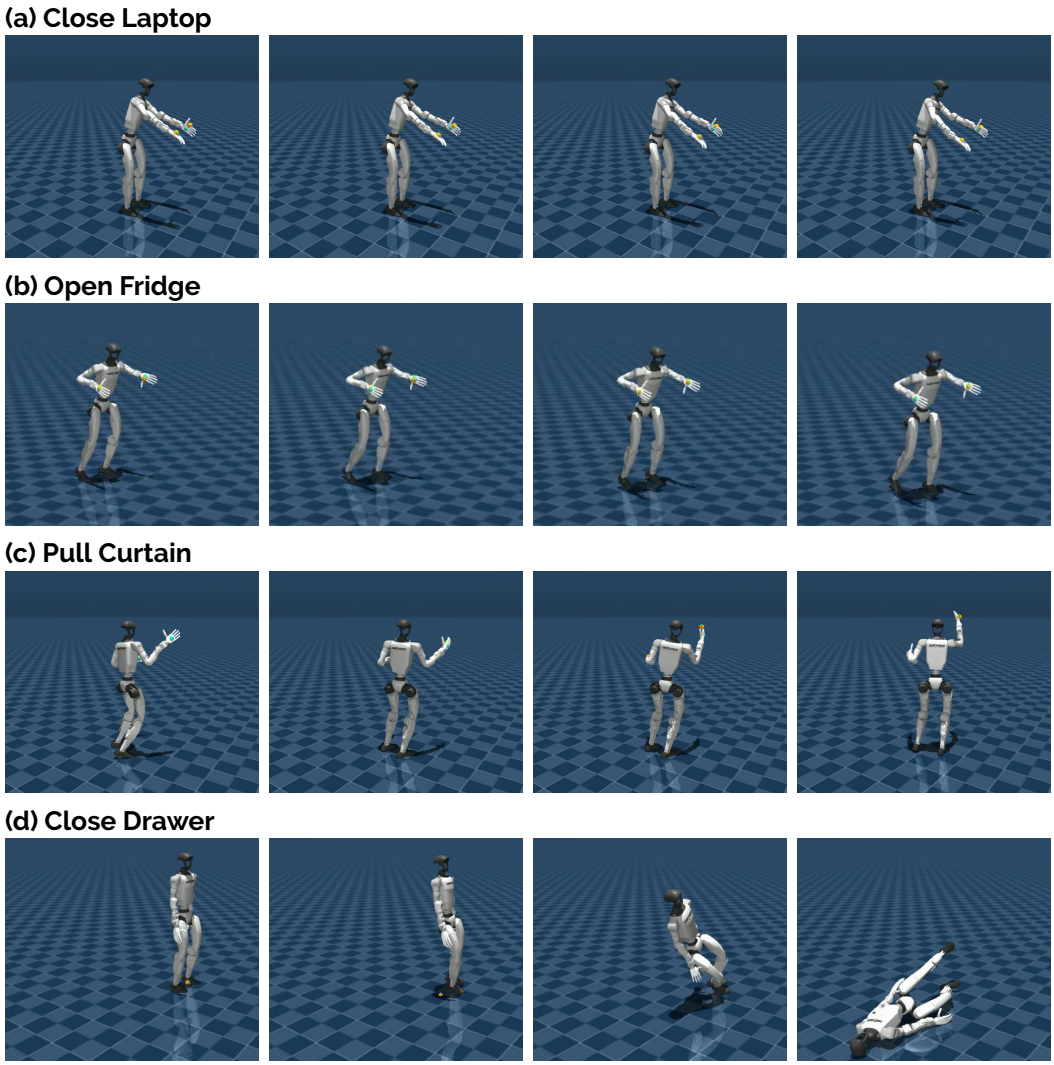



Figure 10: **Online-IK instability: variant B (IK with SONIC balance overlay).** Each row follows one continuous segment from left to right, selected by peak base tilt as described in Appendix D.3. A fixed crop within each sequence preserves relative motion and keeps robot scale comparable across tasks. Close Drawer ends in an initialization fall. The visualizations show recorded robot states without task objects; they do not depict autonomous policy evaluation.

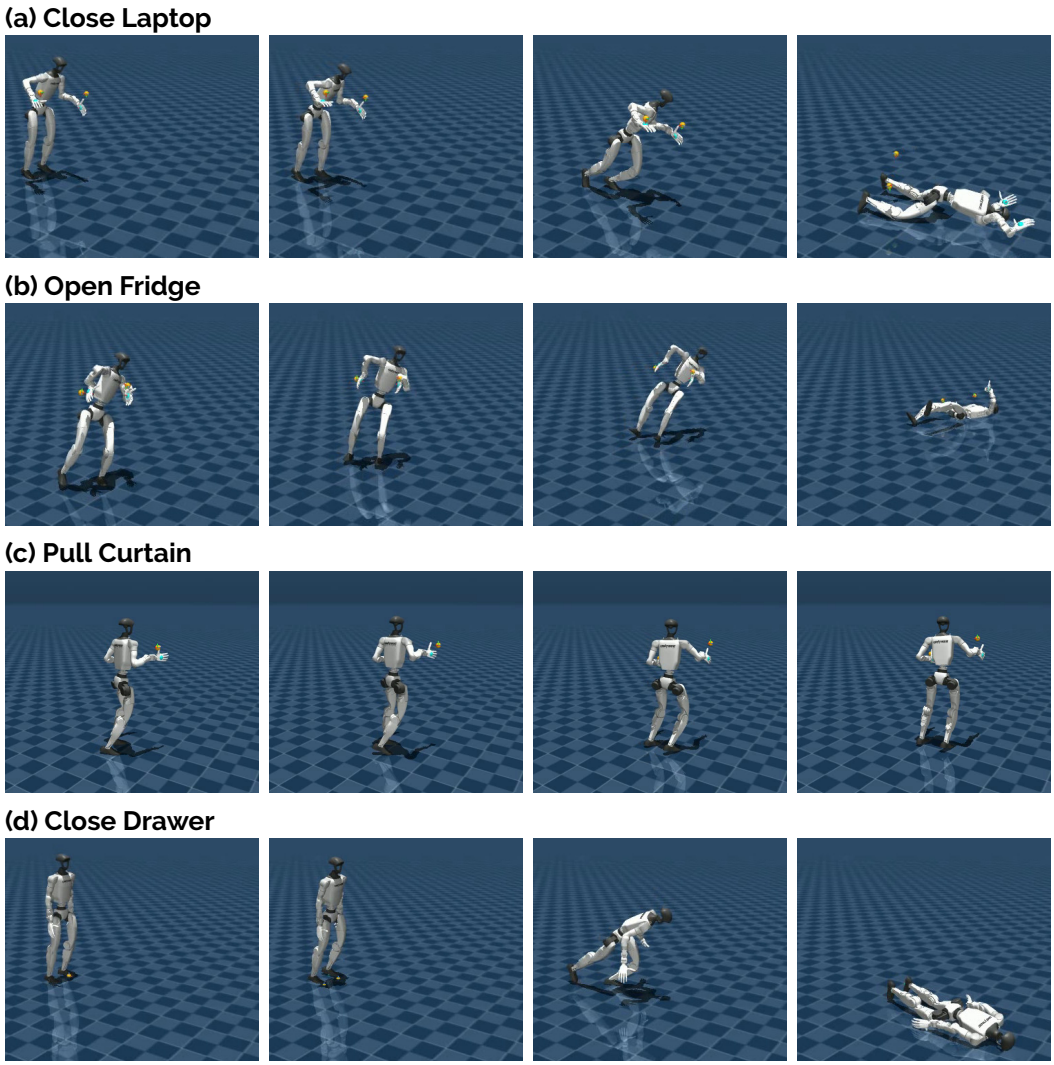



Figure 11: **Online-IK instability: variant C (restricted IK joint control).** Each row follows one continuous segment from left to right, selected by peak base tilt as described in Appendix D.3. A fixed crop within each sequence preserves relative motion and keeps robot scale comparable across tasks. Close Drawer ends in an initialization fall. The visualizations show recorded robot states without task objects; they do not depict autonomous policy evaluation.